$R1 := 168 \qquad R2 := 278 \qquad j := \sqrt{-1}$

$GMR1 := .0313 \qquad GMR2 := .0244$

$Dab := 7 \qquad Dbc := 4.5 \qquad Dca := 2.5$

$$Zii := R1 + .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{GMR2}\right) + 7.93402 \right)$$

$$Zab := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dab}\right) + 7.93402 \right)$$

$$Zbc := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dbc}\right) + 7.93402 \right)$$

$$Zca := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dca}\right) + 7.93402 \right)$$

$$Dan := \sqrt{3.5^2 + 4^2}$$

$$Dcn := \sqrt{4.5^2 + 4^2}$$

$$Dbn := \sqrt{3^2 + 4^2}$$

$$Zan := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dan}\right) + 7.93402 \right)$$

$$Zbn := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dbn}\right) + 7.93402 \right)$$

$$Zcn := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dcn}\right) + 7.93402 \right)$$

$$Zabc := \begin{bmatrix} Zii & Zab & Zca \\ Zab & Zii & Zbc \\ Zca & Zab & Zii \end{bmatrix} \qquad As := 1\angle 120\ deg$$

$$A_s := \begin{bmatrix} 1 & 1 & 1 \\ 1 & As^2 & As \\ 1 & As & As^2 \end{bmatrix}$$

$$Z012 := A_s^{-1} \cdot Zabc \cdot A_s$$

Problem 1 A

$$Zabc = \begin{bmatrix} 168.095 + 1.413i & 0.095 + 0.727i & 0.095 + 0.852i \\ 0.095 + 0.727i & 168.095 + 1.413i & 0.095 + 0.78i \\ 0.095 + 0.852i & 0.095 + 0.727i & 168.095 + 1.413i \end{bmatrix}$$

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Problem 1 B.

$$Z012 = \begin{bmatrix} 168.286+2.968i & -0.052+0.012i & 0.052+0.012i \\ 0.021+0.012i & 168.015+0.636i & 0.072-0.024i \\ -0.021+0.012i & -0.072-0.024i & 167.985+0.636i \end{bmatrix}$$

Problem 1 C.

$Neut := Z012$

$$Neut = \begin{bmatrix} 168.286+2.968i & -0.052+0.012i & 0.052+0.012i \\ 0.021+0.012i & 168.015+0.636i & 0.072-0.024i \\ -0.021+0.012i & -0.072-0.024i & 167.985+0.636i \end{bmatrix}$$

Problem 2

$Dab2 := \sqrt{4^2+2^2}$ $\quad Dbc2 := \sqrt{4^2+2^2}$ $\quad Dac2 := 4$

$Dan2 := \sqrt{6^2+4^2}$ $\quad Dbn2 := 4$ $\quad Dcn2 := \sqrt{2^2+2^2}$

$GMR_2 := .0171$ $\quad r2 := .41$

$$Zii2 := r2 + .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{GMR_2}\right) + 7.93402\right)$$

$$Zab2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dab2}\right) + 7.93402\right)$$

$$Zbc2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dbc2}\right) + 7.93402\right)$$

$$Zac2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dac2}\right) + 7.93402\right)$$

Neutral Impedances

$$Zan2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dan2}\right) + 7.93402\right)$$

$$Zbn2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dbn2}\right) + 7.93402\right)$$

$$Zcn2 := .09530 + j \cdot .12134 \left(\ln\left(\frac{1}{Dcn2}\right) + 7.93402\right)$$

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Phase Impedance Matrice

$$Zabc2 := \begin{bmatrix} Zii & Zab & Zca \\ Zab & Zii & Zbc \\ Zca & Zab & Zii \end{bmatrix}$$

$$Z012_2 := A_s^{-1} \cdot Zabc2 \cdot A_s$$

#2 Answers

$$Zabc2 = \begin{bmatrix} 168.095 + 1.413i & 0.095 + 0.727i & 0.095 + 0.852i \\ 0.095 + 0.727i & 168.095 + 1.413i & 0.095 + 0.78i \\ 0.095 + 0.852i & 0.095 + 0.727i & 168.095 + 1.413i \end{bmatrix}$$

$$Z012_2 = \begin{bmatrix} 168.286 + 2.968i & -0.052 + 0.012i & 0.052 + 0.012i \\ 0.021 + 0.012i & 168.015 + 0.636i & 0.072 - 0.024i \\ -0.021 + 0.012i & -0.072 - 0.024i & 167.985 + 0.636i \end{bmatrix}$$

#3

$$Z012_2 \cdot 5.28 = \begin{bmatrix} 888.55 + 15.669i & -0.272 + 0.063i & 0.272 + 0.063i \\ 0.109 + 0.063i & 887.122 + 3.359i & 0.381 - 0.126i \\ -0.109 + 0.063i & -0.381 - 0.126i & 886.958 + 3.359i \end{bmatrix}$$

Positive, negative, and zero
sequence are the Diagonal
Terms in Matrice.
Z11,Z22,Z33

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